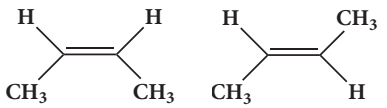
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the opposite side, it is called a Trans isomer.

for example: the geometrical isomers of 2-Butane is expressed as follows



| Cis-2-butene | Trans-2-butene |
|---|---|

## Evaluate Yourself

10) Determine whether each of the following alkenes can exist in cis-trans isomers?

(a) 1 – Chloro propene

(b) 2 – Chloro propene

11) Draw cis-trans isomers for the following compounds

(a) 2- chloro – 2 – butene

(b) $CH_3 - CCl = CH- CH_2CH_3$

### 13.3.1: General methods of preparation of alkenes:

### (1) Preparation of alkene by dehydration of alcohol:

When an alcohol is heated at 430-440 K with excess of concentrated sulphuric acid, a molecule of water from alcohol is removed and an alkene is formed. This reaction is called elimination reaction.

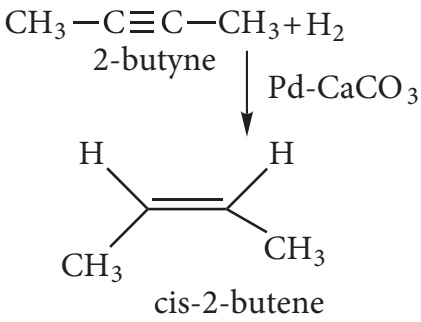
$$\underset{\text{ethanol}}{C_2H_5OH} \xrightarrow[430\text{ -}440K]{\text{Conc. } H_2SO_4} \underset{\text{ethene}}{CH_2{=}CH_2}$$

Ethene can also be prepared in laboratory by catalytic dehydration of alcohol.

$$\underset{\text{ethanol}}{C_2H_5OH} \xrightarrow[623K\text{-}723K]{Al_2O_3} \underset{\text{ethene}}{CH_2{=}CH_2}$$

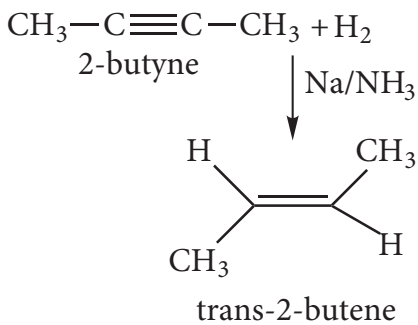
### (2)Preparation of alkenes from alkynes:

Alkynes can be reduced to cis-alkenes using Lindlar's catalyst ($CaCO_3$ supported in palladuium partially deactivated with sulphur (or) gasoline). This reaction is stero specific giving only the cis- alkene.




cis-2-butene

Alkynes can also be reduced to trans-alkenes using sodium in liquid ammonia. This reaction is stereospecific giving only the trans-alkene.




trans-2-butene

### (3) Preparation of alkenes by dehydrohalogenaton of halo alkanes.

Halo alkanes react with alcoholic KOH and eliminate hydrohalide resulting in the formation of alkene.